

Large Scale Protein Extraction

Overview

After establishing suitable conditions on a small scale, scale up the bacterial culture to produce a larger amount of the target protein.

Harvest the supernatant

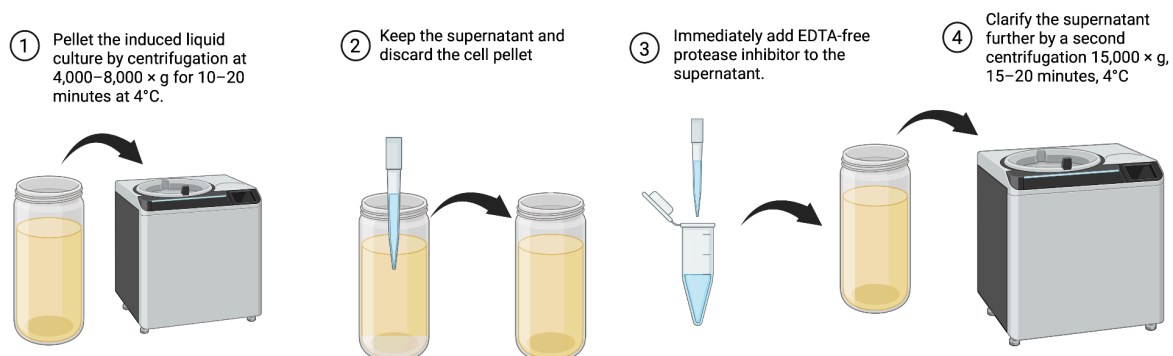
A. Materials

Item	Amount
Liquid culture of <i>B. subtilis</i>	100 mL - 500mL
Centrifuge tubes	Sized to culture volume
Centrifuge	1
EDTA-free protease inhibitor	As required (follow product dilution)
Fresh clean tubes for supernatant	As required
Ice bucket / 4°C access	As required

B. Procedure

1. Pellet the induced liquid culture by centrifugation at 4,000-8,000 × g for 10-20 minutes at 4°C.
2. Keep the supernatant and discard the cell pellet
3. Immediately add EDTA-free protease inhibitor to the supernatant.
4. Clarify the supernatant further by a second centrifugation (e.g., 15,000 × g, 15–20 minutes, 4°C)

Large Scale Extraction



Harvest the cell pellet

A. Materials

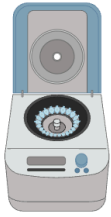
Item	Amount
Liquid culture of <i>B. subtilis</i>	Small scale, 5–50 mL
Centrifuge tubes	Sized to culture volume
Centrifuge	1
CellLytic B 10X concentrate	10-fold
EDTA-free protease inhibitor	As required (follow product dilution)
Lysozyme (10 mg/mL stock)	20 μ L per 1 mL lysis buffer (scale to 0.4 mL: 8 μ L)
Protease inhibitor (100X stock, EDTA-free)	10 μ L per 1 mL lysis buffer (scale to 0.4 mL: 4 μ L)
DNase I (20 mg/mL stock $\rightarrow$ dilute 1:100 to 200 μ g/mL working stock)	25 μ L of working stock per 1 mL lysis buffer (scale to 0.4 mL: 10 μ L)
Wet cell paste	1g = 250 mL culture, OD600 ~2.0
Fresh clean tubes for supernatant	As required
Ice bucket / 4°C access	As required
Vortex	1

B. Procedure

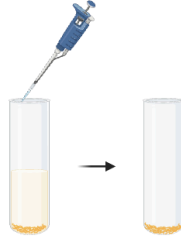
1. Harvest cells by centrifuging at 5,000 g for 10 minutes.
2. Remove the spent medium from the pellet
3. Prepare one 10-fold dilution of CellLytic B 10X into the chosen buffer
4. Add the diluted CellLytic B at 10-20 mL per gram of cell paste, pre-mixed with lysozyme and protease inhibitor at the ratios above. Mix well to fully resuspend the pellet.
5. Incubate with shaking at room temperature for 10–15 minutes.
6. Add DNase I working stock at the ratio above and mix briefly to reduce viscosity before centrifuging.
7. Centrifuge at 16,000 g for 10 minutes to pellet insoluble material.
8. Carefully collect the supernatant (soluble protein fraction).
9. Analyze supernatant and pellet by SDS-PAGE/Western blot to confirm target protein's location.

Harvesting Cell Pellet

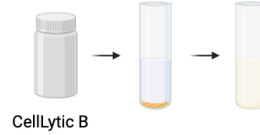
- ① Harvest cells by centrifuging the liquid culture.



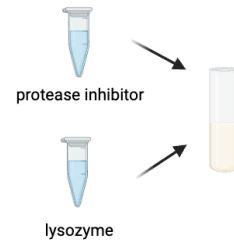
- ② Remove the spent medium and keep the cell pellet.



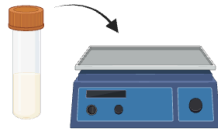
- ③ Resuspend the pellet in diluted CellLytic B.



- ④ Add protease inhibitor and lysozyme.



- ⑤ Incubate with shaking at room temperature for 10-15 minutes then centrifuge.



- ⑥ Add DNase I, mix briefly, then centrifuge at 16,000 g for 10 minutes to clarify.



- ⑦ Collect the supernatant and analyze it alongside with the pellet by SDS-PAGE/Western Blot

